

HW6

$$1. \quad I_{D,sat} = K_n (V_{GS} - V_{tn})^2$$

$$\frac{390}{140} = \left(\frac{4 - V_{tn}}{3 - V_{tn}} \right)^2 \Rightarrow V_{tn} = 1.5V$$

$$390 = K_n (4 - 1.5)^2 \Rightarrow K_n \approx 62 \mu A/V^2$$

$$\frac{W}{L} = 2 \frac{K_n}{K_n'} = 2 \times \frac{62}{100} = 1.24$$

$$2. \quad I_D = \frac{1}{2} K_n (V_{GS} - V_{tn})^2$$

$$0.25 = \frac{1}{2} 200 \times (V_{GS} - 0.6)^2 \Rightarrow V_{GS} = 1.718V$$

$$V_S = -1.718V \quad V_D = 9V - 0.25 \times 24k = 3V$$

saturation region

$$3. (a) \quad V_G = 10 \times \frac{1000}{100+3200} = 3.125V$$

$$K_n = K_n' \frac{W}{L} = 100 \times 6 = 600$$

$$I_D = \frac{1}{2} \times 600 \times (V_{GS} - 0.75)^2$$

$$= 300 \times 10^{-6} (3.125 - 240000 I_D - 0.75)^2$$

$$I_D = 9.17 \mu A$$

$$V_{GS} = 3.125 - 9.17 \mu A \times 240k\Omega = 0.925$$

$$V_{DS} = 10 - 9.17 \mu A \times 120k\Omega - 2.200 = 6.7V$$

$$(V_{GS}, V_{DS}, I_D) = (0.925V, 6.7V, 9.17 \mu A)$$

$$(b) \quad I_D = \frac{1}{2} \times 1200 \times (3.125 - 240000 I_D - 0.75)^2$$

$$I_D = 9.375 \mu A$$

$$V_{GS} = 3.125 - 9.375 \mu A \times 240k = 0.875V$$

$$V_{DS} = 10 - 9.375 \mu A \times 120k - 2.250 = 6.625V$$

$$(V_{GS}, V_{DS}, I_D) \approx (0.875V, 6.625V, 9.375 \mu A)$$

$$4. (a) \quad I = 0.75 (1.6 - V_1)^2$$

$$I = 0.75 (-V_2 - 0.9)^2$$

$$x = 1.6 - V_1 = -V_2 - 0.9$$

$$V_1 = 1.6 - x \quad V_2 = -0.9 - x$$

$$I = V_2 + 2.5 = 1.6 - x \quad 0.75x^2 = 1.6 - x$$

$$x = 0.939 \Rightarrow V_1 = 0.661 \quad V_2 = -1.839$$

$$(b) \quad y = V_{GS1} = V_{GS2} \quad V_3 - V_4 = y \quad V_4 - V_5 = y$$

$$V_3 = V_5 + 2y \quad 5 - V_3 = V_5 \quad V_5 = 2.5 - y$$

$$I = V_5 = 2.5 - y \quad I = 0.75 (y - 0.9)^2$$

$$0.75 (y - 0.9)^2 = 2.5 - y \quad y = 1.839$$

$$\Rightarrow V_5 = 0.661V \quad V_4 = 2.5V \quad V_3 = 4.339V$$

$$5. (a) \quad I_D = 4(0.2)^2 = 0.16mA$$

$$R_D = \frac{1 - 0.2}{0.16mA} = 5k\Omega$$

$$R_S = \frac{-0.6 + 1}{0.16mA} = 2.5k\Omega$$

$$(b) \quad V_D = V_S + V_{ov} = -0.6 + 0.2 = -0.4$$

$$R_{D,max} = \frac{1 - (-0.4)}{0.16mA} = 8.75k\Omega$$

$$b. \quad V_G = \frac{510}{510+510} \times 15 = 7.5V$$

$$V_S = 15 - 100I_D$$

$$V_D = 50I_D$$

$$I_D = \frac{1}{2} 0.8 \times (V_{GS} - 0.75)^2$$

$$= 0.4 \times (6.75 - 100I_D)^2$$

$$I_D = 63.5 \mu A$$

$$V_S = 15 - 100(0.0635) = 8.65V$$

$$V_{GS} = 7.5 - V_S = -1.15V$$

$$V_{DS} = 3.18 - 8.65 = -5.47V$$